Signed-collision preflight source review

5 November 2026

Scoped source fidelity supported. Published `c531856dcb72`; candidate `d8c496084c1e`. No compiler or runtime execution.

L1 correction. Metadata, prime predicates and canonical nonzero inputs are checked before construction. Coefficient equality and source validity/order checks precede transfer. Zero images return Err afterward. The false opt-in branch preserves original assertion order without new field arithmetic. Full collector tail is byte-identical.

Mathematical scope. Nonzero G, prime r and $rG = O$ imply order r under the existing arithmetic premises. Valid Q alone does not establish membership in the generated subgroup; that remains explicit. Transfer homomorphism and prime-predicate correctness remain dependencies. Collision signs, zero-denominator cases and final source equality are unchanged.

Bounds and costs. $p < 2^{31}$ gives $p^2 < 2^{62}$. Canonical residues below $r \leq 2^{63}$ add within unsigned64; products use unsigned128. New prime/comparison work has no measured price outside field counters. Validity/order work moves into a named preflight phase. Err lacks phase totals, requiring external complete failure receipts.

Cost boundary. The phase mixes curve, generator and order checks with target-dependent Q validation. Its name defines neither online nor reusable cost. A future measured protocol must split these costs explicitly; unchanged source does not prove runtime equivalence.

Remaining boundaries. Constructor, transfer and later arithmetic, observers, allocations and counters can still panic. No total Result, timing, memory or complete-cost claim is established. Manifest and published bytes match; source-only qualification remains distinct from future native execution.